Information Retrieval

Lecture 6

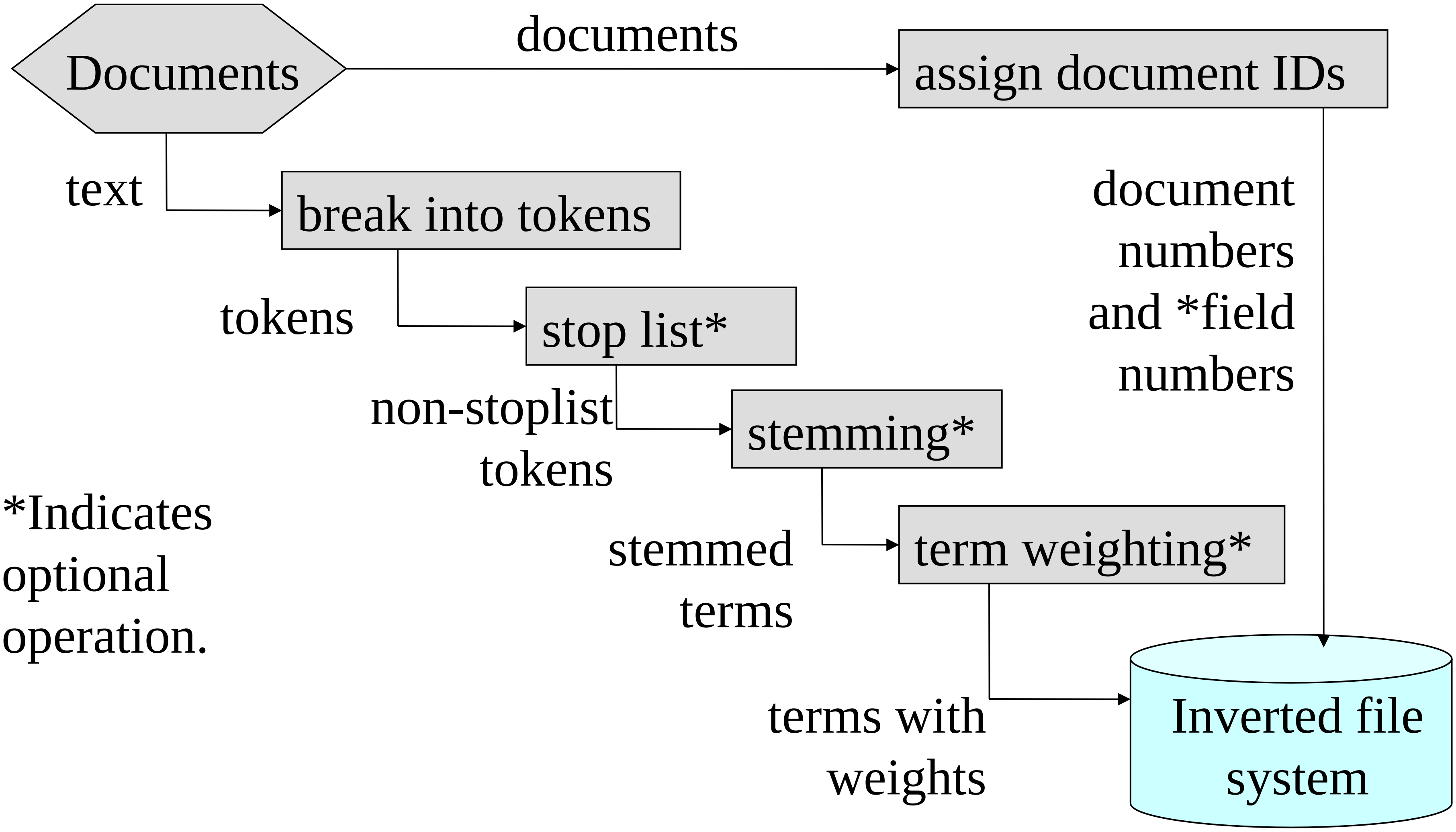
Searching

SMART System

An experimental system for automatic information retrieval

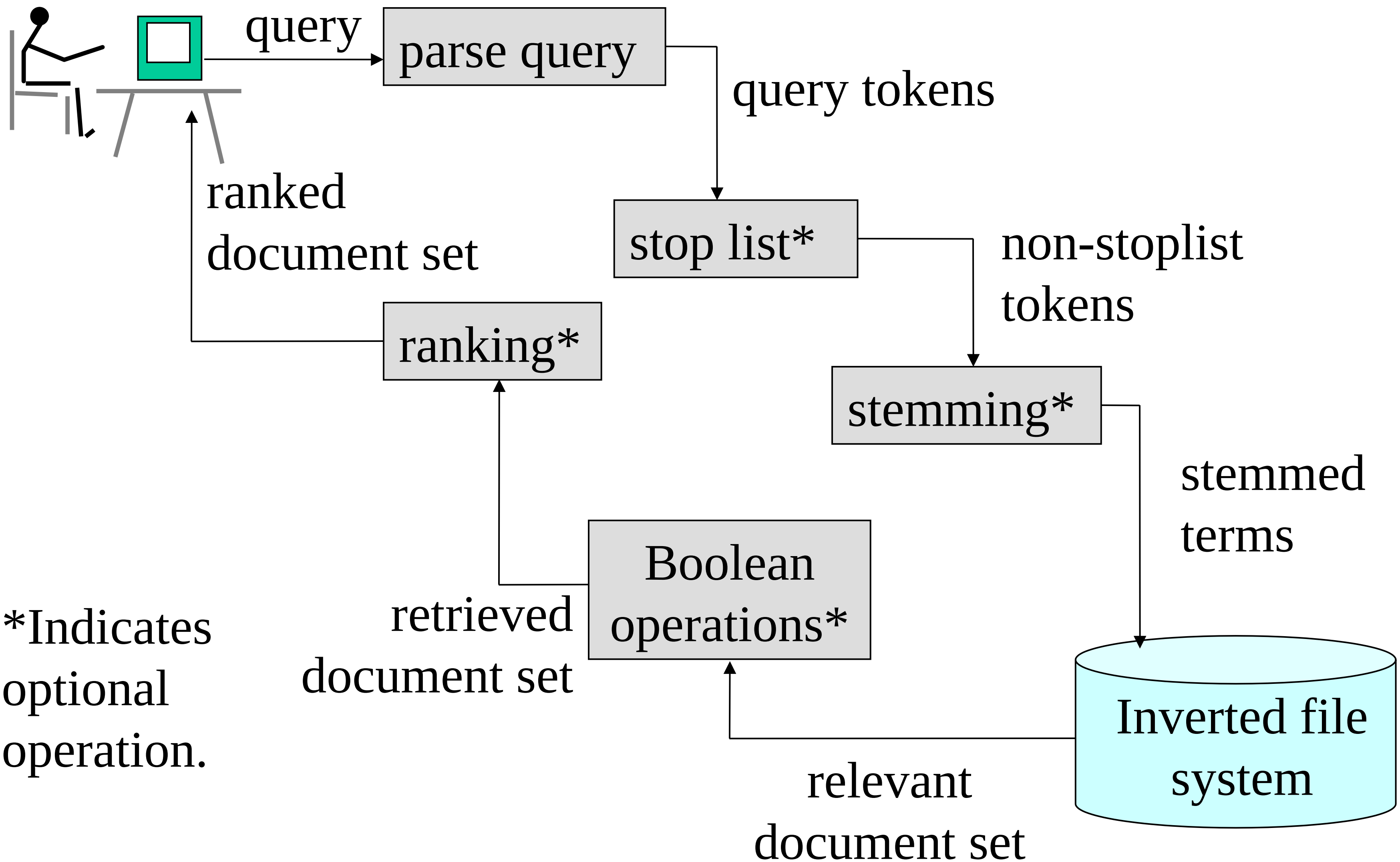
- automatic indexing to assign terms to documents and queries
- collect related documents into common subject classes
- identify documents to be retrieved by calculating similarities between documents and queries
- procedures for producing an improved search query based on information obtained from earlier searches

Indexing Subsystem



*Indicates optional operation.

Search Subsystem



*Indicates optional operation.

Decisions in Building the Word List: What is a Term?

- Underlying character set, e.g., printable ASCII, Unicode, UTF8.
- Is there a controlled vocabulary? If so, what words are included?
- List of stopwords.
- Rules to decide the beginning and end of words, e.g., spaces or punctuation.
- Character sequences not to be indexed, e.g., sequences of numbers.

Lexical Analysis: Term

What is a term?

Free text indexing

A term is a group of characters, extracted from the input string, that has some collective significance, e.g., a complete word.

Usually, terms are strings of letters, digits or other specified characters, separated by punctuation, spaces, etc.

Lexical Analysis: Choices

Punctuation: In technical contexts, punctuation may be used as a character within a term, e.g., *wordlist.txt*.

Case: Case of letters is usually not significant.

Hyphens:

(a) Treat as separators: *state-of-art* is treated as *state of art*.

(b) Ignore: *on-line* is treated as *online*.

(c) Retain: *Knuth-Morris-Pratt Algorithm* is unchanged.

Digits: Most numbers do not make good terms, but some are parts of proper nouns or technical terms: *CS430*, *Opus 22*.

Lexical Analysis: Choices

The modern tendency, for ***free text searching***, is to map upper and lower case letters together in index terms, but otherwise to minimize the changes made at the lexical analysis stage.

With ***controlled vocabulary***, the lexical decisions are made in creating the vocabulary.

Lexical Analysis Example: Query Analyzer

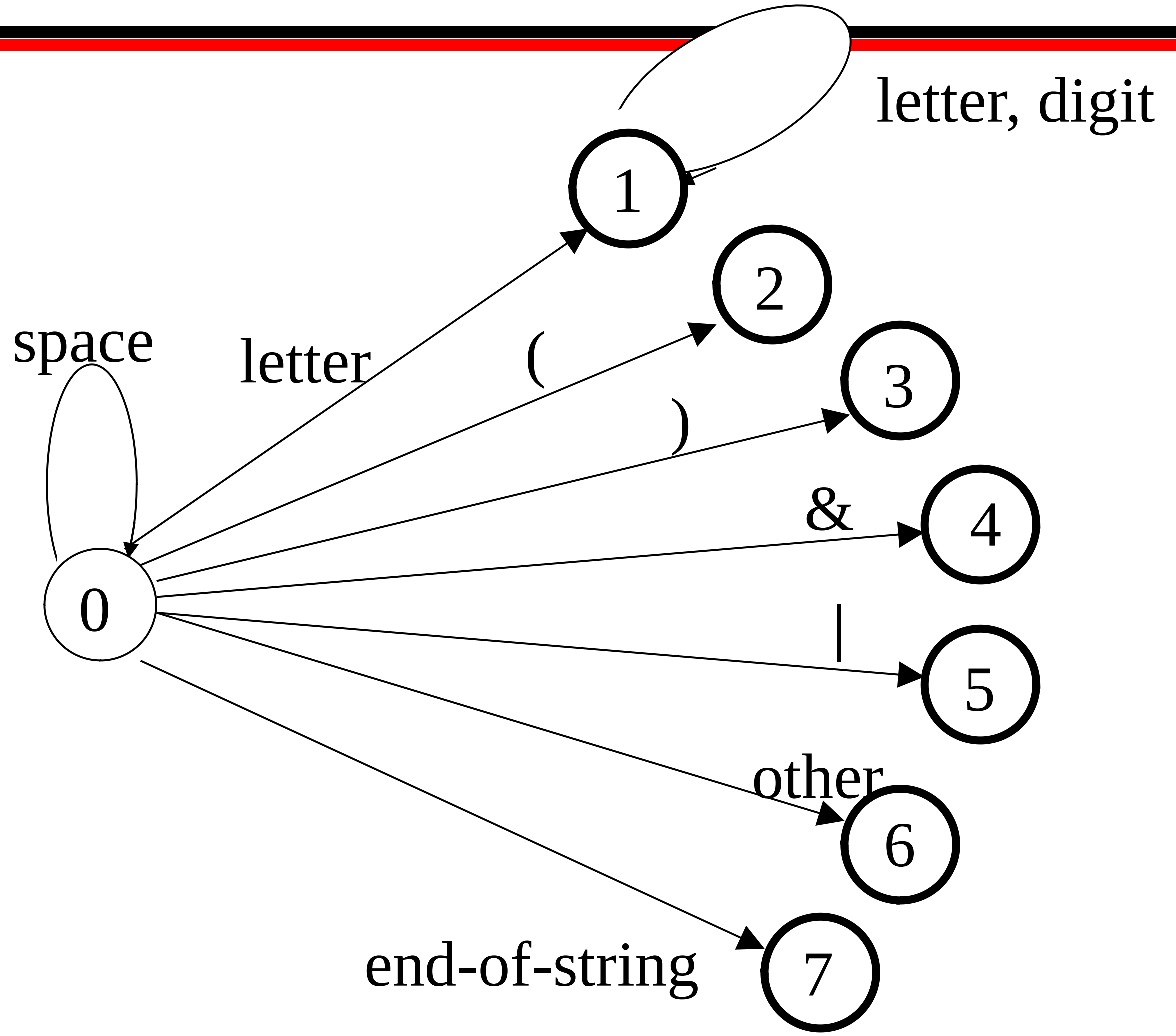
A **term** is a letter followed by a sequence of letters and digits.

Upper case letters are mapped into the lower case equivalents.

The following characters have significance as **operators**:

( ) & |

Lexical Analysis: Transition Diagram



Lexical Analysis: Transition Table

State	space	letter	(	)	&	|	other	end-of string	digit
0	0	1	**2**	**3**	**4**	**5**	**6**	**7**	**6**
1	**1**	1	**1**	**1**	**1**	**1**	**1**	**1**	1

States in red are final states.

Changing the Lexical Analyzer

This use of a transition table allows the system administrator to establish differ lexical choices for different collections of documents.

Example:

To change the lexical analyzer to accept tokens that begin with a digit, change the top right element of the table to 1.

Stop Lists

Very common words, such as *of*, *and*, *the*, are rarely of use in information retrieval.

A **stop list** is a list of such words that are removed during lexical analysis.

A long stop list saves space in indexes, speeds processing, and eliminates many false hits.

However, common words are sometimes significant in information retrieval, which is an argument for a short stop list. (Consider the query, "To be or not to be?")

Example: Stop List for Assignment 1

a	about	an	and
are	as	at	be
but	by	for	from
has	have	he	his
in	is	it	its
more	new	of	on
one	or	said	say
that	the	their	they
this	to	was	who
which	will	with	you

Example: the WAIS stop list (first 84 of 363 multi-letter words)

about	above	according	across	actually	adj
after	afterwards	again	against	all	almost
alone	along	already	also	although	always
among	amongst	an	another	any	anyhow
anyone	anything	anywhere	are	aren't	around
at		be	became	because	become
becomes	becoming	been	before	beforehand	begin
beginning	behind	being	below	beside	besides
between	beyond		billion	both	but by
can	can't		cannot	caption	co
could	couldn't				
did	didn't	do	does	doesn't	don't
down	during	each	eg	eight	eighty
either	else	elsewhere	end	ending	enough
etc	even	ever	every	everyone	everything

Suggestions for Including Words in a Stop List

- **Include** the most common words in the English language (perhaps 50 to 250 words).
- **Do not include** words that might be important for retrieval (Among the 200 most frequently occurring words in general literature in English are *time*, *war*, *home*, *life*, *water*, and *world*).
- In addition, **include** words that are very common in context (e.g., *computer*, *information*, *system* in a set of computing documents).

Stop list policies

How many words should be in the stop list?

- Long list lowers recall

Multi-lingual document collections have special problems, e.g., *die* is a very common word in German but less common in English.

There is very little systematic evidence to use in selecting a stop list.

Stop Lists in Practice

The modern tendency is:

a) have very short stop lists for broad-ranging or multi-lingual document collections, especially when the users are not trained.

b) have longer stop lists for document collections in well-defined fields, especially when the users are trained professional.

Stemming

Morphological variants of a word (morphemes). Similar terms derived from a common stem:

engineer, engineered, engineering
use, user, users, used, using

Stemming in Information Retrieval. Grouping words with a common stem together.

For example, a search on *reads*, also finds *read*, *reading*, and *readable*

Stemming consists of removing **suffixes** and **conflating** the resulting morphemes. Occasionally, **prefixes** are also removed.